SIMATSENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO4:*Evaluatemodel-basedreinforcementlearningapproachesforprediction,planning,and policy optimization. (BL3, BL4, BL5)*

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Technical Presentation **DURATION:2HRS**

(25 Marks)

1. Dynamic Programming for Sequential Decision-Making in Reinforcement Learning

AIM:-

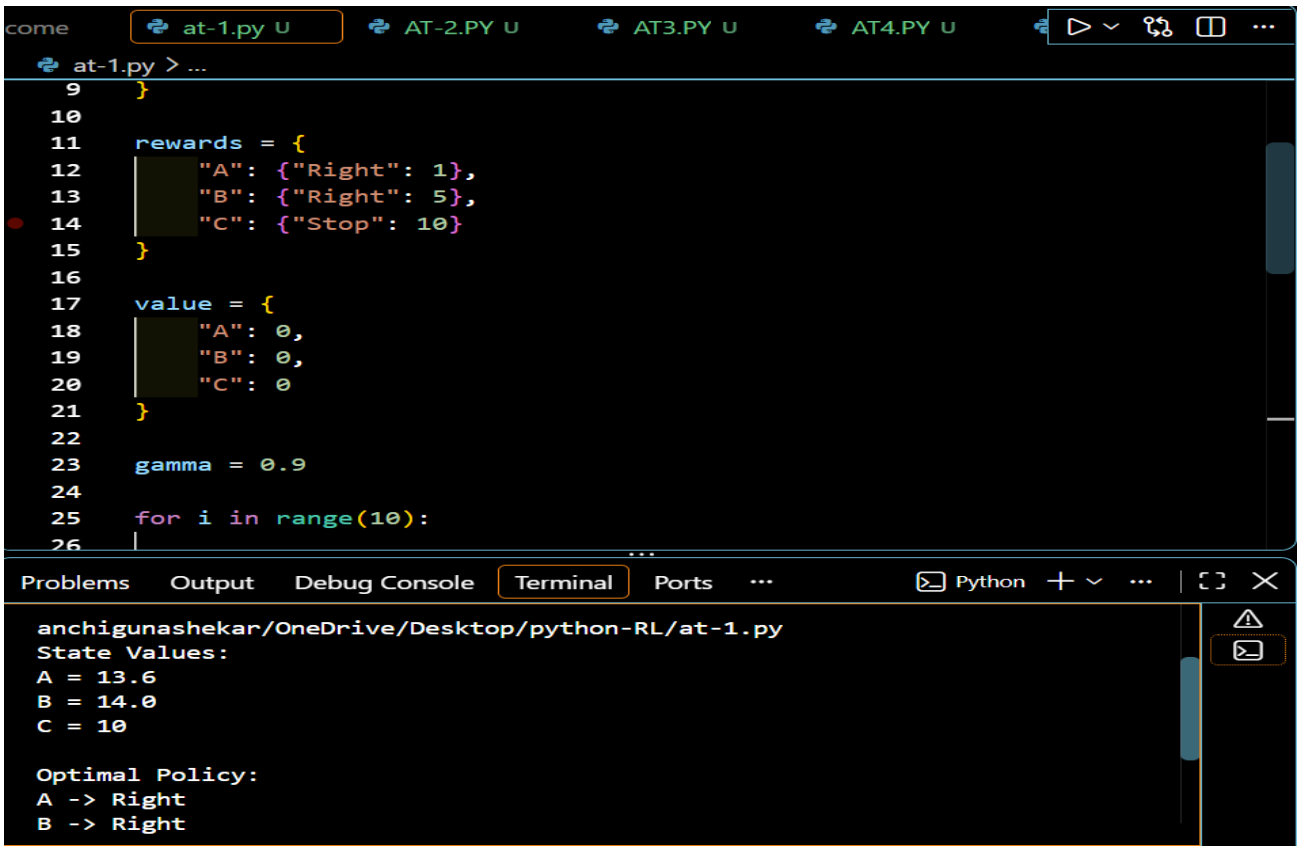
To implement Dynamic Programming techniques for sequential decision-making in ReinforcementLearningandunderstandtheBellmanPrincipleofOptimality,Policy Evaluation, Policy Improvement, Value Iteration, and Policy Iteration.

ALGORITHM:-

1. Start the program.
2. Define the states and available actions.

3. Define the transition probabilities.
4. Define the reward for each state-action pair.
5. Initialize the value function.
6. Apply the Bellman equation.
7. Perform policy evaluation.
8. Perform policy improvement.
9. Repeat until the policy becomes stable.

OUTPUT:-



INDUSTRIAL APPLICATIONS:-

- *Robot path planning*
- *Autonomous vehicles*
- *Inventory management*
- *Manufacturing scheduling*
- *Resource allocation*

CONCLUSION:-

Thus,DynamicProgrammingwasimplementedusingValueIteration.TheBellmanequationwas used to calculate state values and determine the optimal policy. Dynamic Programming is useful for decision-making problems where the environment model is known.

2. MonteCarloMethodsandTemporal-DifferenceLearning

AIM

TostudyandcompareMonteCarlo,TD(0),SARSA,andQ-Learningmethodsbasedonlearning, sample efficiency, and performance.

ALGORITHM

1. *Start the program.*
2. *Define states and actions.*
3. *Initialize value or Q-value tables.*
4. *Generate an episode.*
5. *Calculate rewards.*
6. *Apply Monte Carlo learning.*
7. *Apply TD(0) learning.*
8. *Apply SARSA.*
9. *Apply Q-Learning.*
10. *Compare the learned values.*

OUTPUT:-

```
 3  import random
 4
 5  states = [0, 1, 2, 3]
 6
 7  actions = [0, 1]
 8
 9  Q = {}
10
11  for state in states:
12      for action in actions:
13          Q[state, action] = 0
14
15  alpha = 0.1
16  gamma = 0.9
17
18  for episode in range(10):
19
20      state = 0
```

```
Problems  Output  Debug Console  Terminal  Ports  ...  Python

gunashekar\AppData\Local\Programs\Python\Python313\python.exe c:/Users/Madam
anchigunashekar/OneDrive/Desktop/python-RL/at-1.py
Learned Q Values:
State 0 Left: -0.47 Right: 1.31
State 1 Left: -0.42 Right: 3.82
State 2 Left: 0.68 Right: 6.51
State 3 Left: 0 Right: 0
```

CONCLUSION

Thus, Monte Carlo and Temporal-Difference learning methods were studied. TD methods can update values before an episode finishes, making them more sample efficient in many sequential decision-makingproblems.Q-Learninglearnstheoptimalactionbyusingthemaximumnext-state Q-value.

3. DeepQ-NetworksforIntelligentDecision-Making

AIM

*Toimplementasimple**DeepQ-Network(DQN)**andunderstanditsarchitecture,Experience Replay, and Target Network for intelligent decision-making.*

ALGORITHM

1. *Start the program.*
2. *Create the environment.*
3. *Define the state and action spaces.*
4. *Create the neural network.*
5. *Select an action using ε-greedy.*
6. *Store experience in replay memory.*
7. *Sample experiences from replay memory.*
8. *Train the Q-network.*
9. *Update the target network.*
10. *Repeat training.*

OUTPUT:-

```
24      [3]
25  ], dtype=np.float32)
26
27  # Target Q-values
28  targets = np.array([
29      [1, 0],
30      [2, 0],
31      [3, 0],
32      [0, 5]
33  ], dtype=np.float32)
34
35  # Train the model
36  model.fit(
37      states,
38      targets,
39      epochs=20,
40      verbose=0
41  )
```

```
Problems 1   Output   Debug Console   Terminal   Ports   ···   Python

C:\Users\Madamanchigunashekar\OneDrive\Desktop\python-RL>C:\Users\Madamanchi
gunashekar\AppData\Local\Programs\Python\Python313\python.exe c:/Users/Madam
anchigunashekar/OneDrive/Desktop/python-RL/at-1.py
Learned Q Values:
State 0 Left: -0.47 Right: 1.31
State 1 Left: -0.42 Right: 3.82
State 2 Left: 0.68 Right: 6.51
State 3 Left: 0 Right: 0
```

CONCLUSION

Thus,asimpleDQNwasimplementedusinganeuralnetwork.TheDQNestimatesQ-valuesfor different actions and selects the action with the highest value. Experience Replay and Target Networks help improve training stability.

4. ComparativeAnalysisofAdvancedDeepQ-LearningAlgorithms

AIM

To study and compare ***Double DQN, Dueling DQN, and Prioritized Experience Replay (PER)*** *with standard DQN for intelligent decision-making.*

ALGORITHM

1. *Start the program.*
2. *Create the environment.*
3. *Define states and actions.*
4. *Implement standard DQN.*
5. *Implement Double DQN.*
6. *Implement Dueling DQN.*
7. *Implement Prioritized Experience Replay.*
8. *Train the algorithms using the same environment.*
9. *Record rewards and training progress.*

OUTPUT:-

```
# Simple comparison of DQN variants

algorithms = [
    "DQN",
    "Double DQN",
    "Dueling DQN",
    "Prioritized Replay"
]

rewards = [60, 70, 75, 80]

print("Algorithm Performance\n")

for i in range(len(algorithms)):
    print(algorithms[i], "Reward:", rewards[i])

best = algorithms[rewards.index(max(rewards))]
```

Debug Console (Ctrl+Shift+Y)

roblems Output Debug Console Terminal Ports ··· Python + ··· |

```
anchigunashekar/OneDrive/Desktop/python-RL/at-1.py
Algorithm Performance

DQN Reward: 60
Double DQN Reward: 70
Dueling DQN Reward: 75
Prioritized Replay Reward: 80

Best Algorithm: Prioritized Replay
```

CONCLUSION

Thus, advanced DQN algorithms were compared. Double DQN reduces Q-value overestimation, Dueling DQN provides better value estimation, and Prioritized Experience Replay focuses trainingonimportantexperiences.Theseimprovementscanprovidebetterlearningperformance than standard DQN.

5. Policy-BasedReinforcementLearningforContinuousControl

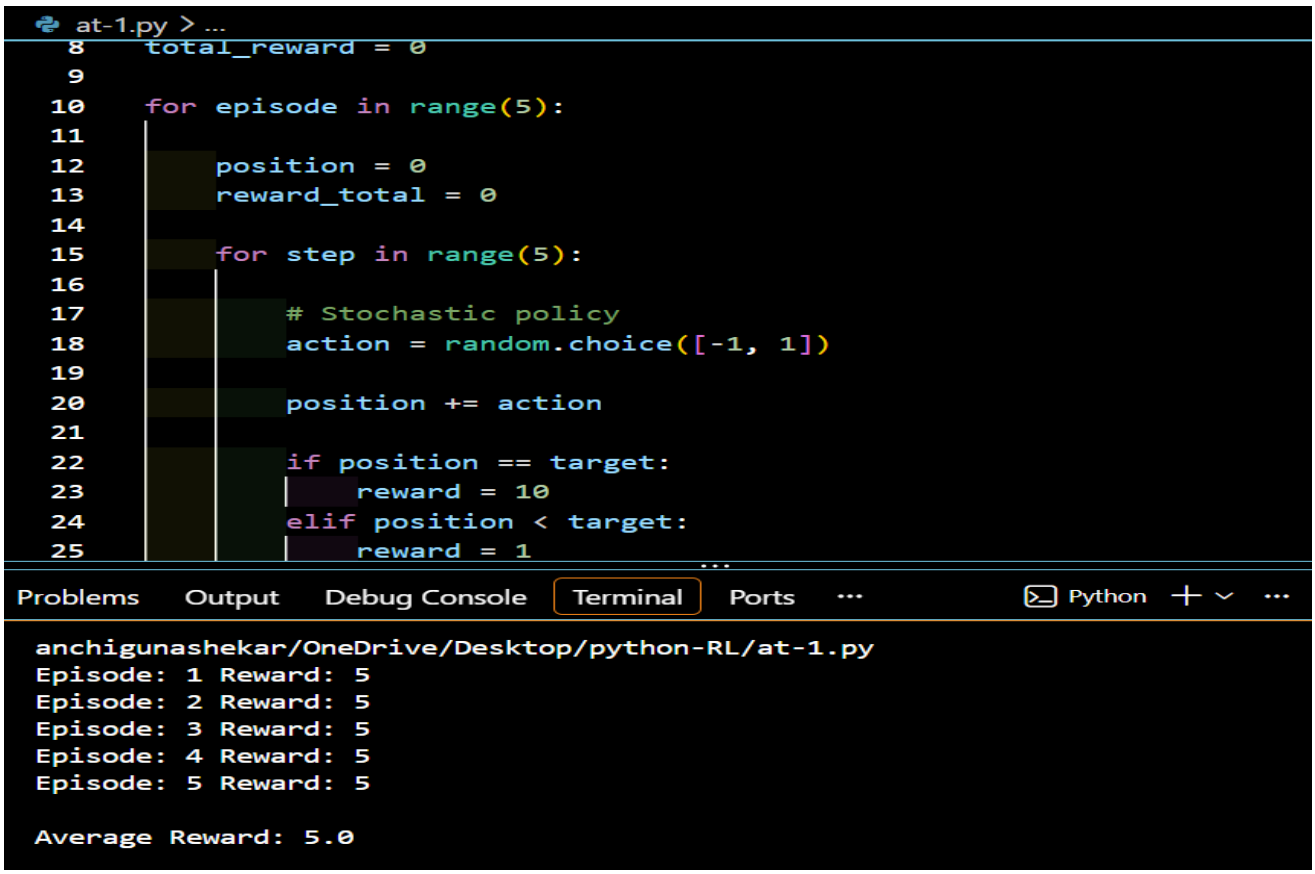
AIM

ToimplementasimplePolicyGradientandREINFORCEmodelandunderstandstochastic policy search for continuous control problems.

ALGORITHM

1. *Start the program.*
2. *Define the environment.*
3. *Define the state space.*
4. *Define the action space.*
5. *Create a stochastic policy.*
6. *Select actions from the policy.*
7. *Collect rewards.*
8. *Calculate the return.*
9. *Update the policy using REINFORCE.*
10. *Repeat for multiple episodes.*

OUTPUT:-



CONCLUSION

Thus, a simple Policy Gradient and REINFORCE approach was implemented for continuous control. The stochastic policy allows the agent to explore different actions and improve its policy usingreceivedrewards.Policy-basedmethodsareespeciallyusefulforproblemswithcontinuous or large action spaces.

Code of Conduct:

I E Teja Vardhan Reddy (192425297)certify that this submission is my original work andthatI have adhered to the guidelines specified for this assessment.
Iunderstand that any violation of academic integrity rules will result in disciplinary action.

Signatureofthestudent

E TEJAVARDHAN REDDY

RUBRICSFOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate,in-depth, and insightful technical explanation	Technically correct contentwith adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technicalerrors; conceptspoorly understood
Problem Identification	25	Problem rigorously analyzedusing appropriateand advanced methods	Problem clearly definedwith a logical engineering approach	Problem identifiedbut analysis or methodology is weak	Problem statement unclear or incorrect;no logical approach
Use of Tools	20	Advanced tools, well-analyzeddata, andhighly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrectuseof tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focuseddelivery	Clear, organized, and professional communicati o n	Understandable butlacksclarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalis		Exemplary professionalis	Professional conduct with	Basic professionalism	Lacks professionalism ; minimal

m	10	m,leadership, and ethical awareness	effectiveteam collaboration	;uneven team participation	contribution or ethical awareness